Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	22 September 2026
Team ID	
Project Name	“An Employee Requests Installation of Licensed Software through the Service Catalog
Maximum Marks	4 Marks

Technical Architecture:

The technical architecture uses ServiceNow Service Portal and Service Catalog as the user-facing layer, Flow Designer/Workflow for process automation, approval and fulfillment components for request processing, and ServiceNow database records for storing request information. This architecture provides a centralized and trackable process from **software request submission to approval, installation, and completion**.

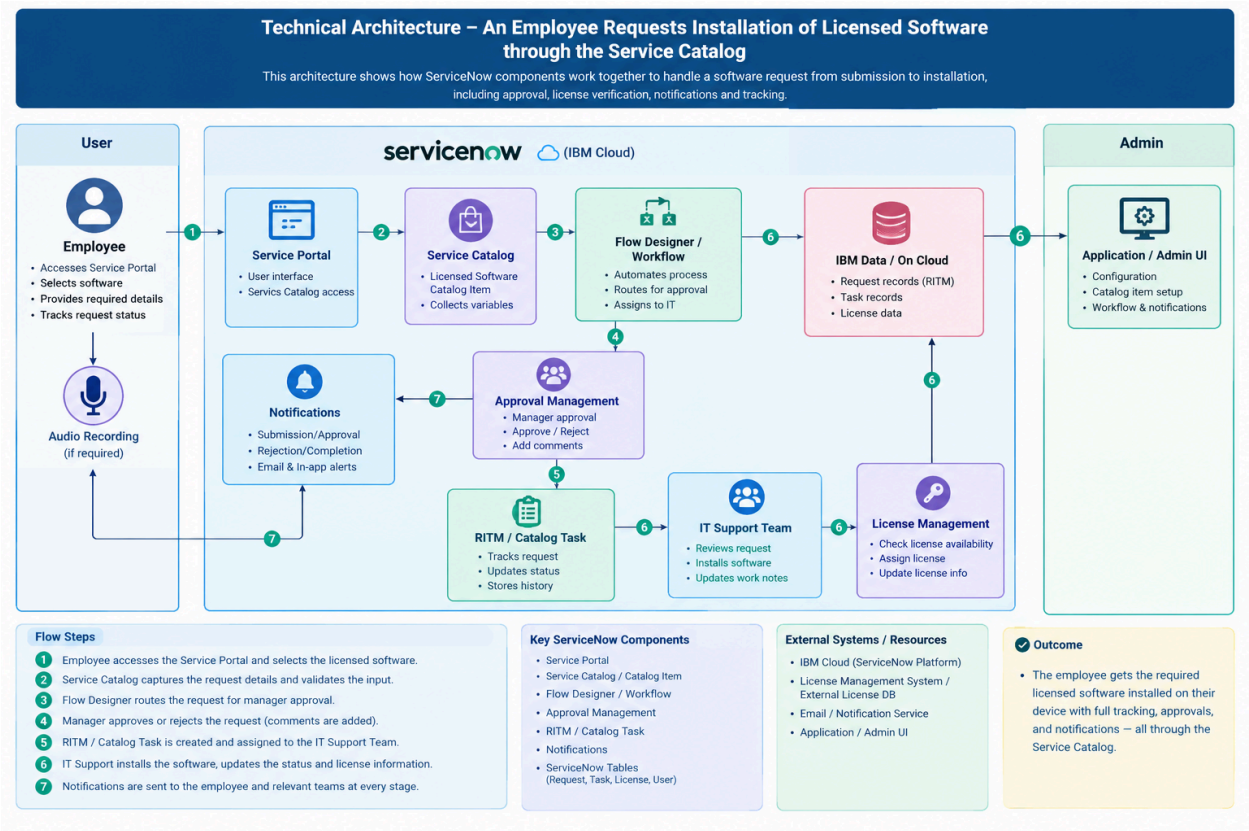


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employee accesses the Service Portal to view the Service Catalog, select licensed software, submit requests, and track request status.	ServiceNow Service Portal / HTML, CSS, JavaScript
2.	Application Logic-1	Handles software request submission, validation of catalog variables, request creation, and request status updates.	ServiceNow Service Catalog / JavaScript
3.	Application Logic-2	Automates the software request workflow, including approval routing, task creation, assignment, and status updates.	ServiceNow Flow Designer / Workflow
4.	Application Logic-3	Manages manager approval or rejection and records approval comments for the software request.	ServiceNow Approval Management
5.	Database	Stores employee details, software requests, request items, catalog tasks, approvals, work notes, and request status information.	ServiceNow Platform Database / ServiceNow Tables
6.	Cloud Database	Maintains software and license-related information when an external or cloud-based license repository is used.	ServiceNow Cloud Database / External License Database
7.	File Storage	Stores supporting documents or attachments related to the software request, if required.	ServiceNow Attachment Storage
8.	External API-1	Can be used to retrieve or verify software license availability from an organization's license management system.	REST API / ServiceNow IntegrationHub
9.	External API-2	Can be used to communicate with an enterprise software deployment or endpoint-management system for software installation.	REST API / IntegrationHub
10	Machine Learning Model	Not required for the core licensed software request process. The project can be implemented without a Machine Learning model.	Not Applicable
11	Infrastructure (Server / Cloud)	Provides the platform where the Service Portal, Service Catalog, workflows, request records, approvals, and notifications are executed.	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The core application is developed on the ServiceNow platform. Custom client-side and server-side logic can be implemented using JavaScript. No separate open-source framework is required for the core solution.	ServiceNow Platform, JavaScript
2.	Security Implementations	Access to software requests is controlled using user roles and permissions. Only authorized employees can submit requests, managers can approve or reject requests, and IT Support users can fulfill assigned requests.	ServiceNow Roles & ACLs, User Authentication, Access Controls
3.	Scalable Architecture	The solution is hosted on the ServiceNow cloud platform and can handle increasing numbers of employees, software requests, approvals, and fulfillment tasks. Additional catalog items and workflows can be added as business requirements grow.	ServiceNow Cloud Platform, Service Catalog, Flow Designer
4.	Availability	The application is available through the ServiceNow platform so employees can access the Service Catalog and track their requests. The cloud-based platform supports continuous service availability without requiring an application server to be maintained by the project team.	ServiceNow Cloud Platform, Service Portal
5.	Performance	The solution uses ServiceNow catalog forms, automated workflows, and server-side processing to handle requests efficiently. Mandatory-field validation and automated routing reduce unnecessary processing and manual communication.	ServiceNow Service Catalog, Flow Designer, JavaScript